



Thermoplastics:

THINNOVATION Is The DESIGN CHALLENGE In Mobile

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We have always talked about how process nodes and newer semiconductors are helping with miniaturisation, but that is not the full story. This interview takes a look at the other side of miniaturisation, of the materials that allow you to build such small-scale objects in the first place. Sridhar N.B., director, sales and marketing (industrial), DSM India Pvt Ltd, speaks with Dilin Anand of EFY

Q. How are materials helping with increased miniaturisation of electronics?

A. Miniaturisation and, linked to that, thinnovation, is the design challenge in mobile. Newer materials that have the strength to be used in thinner structures while maintaining strength results in reduced size of components, while putting more features into these. For example, newer connectors have to deliver more power and signals in a small form factor.

One of the best ways to achieve this is through next-generation thermoplastics that can not only deliver on specifications but also open the door to completely new designs and concepts as we have seen in some recent consumer devices.

Q. Could you give us an example of thinnovation making an impact in electronics?

A. When we talk about thinnovation, over the past few years, thickness of smartphones has

reduced by 12 per cent per year, on average. One of the latest developments to enable this trend is the new-generation universal serial bus (USB) Type-C connectors.

These connectors need to carry more power than previous-generation connectors in a much smaller form factor. So performance properties of the materials used for holding all conductive elements together are especially critical.

Reliability is a key requirement, so the industry is looking for